

23-2

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issuance of changes to
FAR Parts 23 and 25

Title 14—AERONAUTICS AND SPACE

Chapter I—Federal Aviation Agency

[Docket No. 2029; Amdt. 23-2, 25-6]

PART 23—AIRWORTHINESS STANDARDS: NORMAL, UTILITY, AND ACROBATIC CATEGORY AIRPLANES

PART 25—AIRWORTHINESS STANDARDS: TRANSPORT CATEGORY AIRPLANES

Limited Weight Credit for Airplanes Equipped With Standby Power

The purpose of this amendment is to provide a limited increase in the maximum certificated takeoff and landing weights for airplanes equipped with rocket engines used to provide standby power. This action was published as a notice of proposed rule making and circulated as Federal Aviation Notice No. 63-41 (28 F.R. 11481), October 26, 1963.

Standby power is obtained from rocket engines and is separate from the power obtained from the airplanes' main engines. Standby power is available for a relatively short time for use in cases of emergency. Since this power is capable of producing a temporary increase in airplane climb performance, it can be useful in the takeoff and landing regimes of flight where its temporary nature is not a deterrent to its use.

Under currently effective regulations, there are instances when operators of transport category as well as nontransport category airplanes with a standby power rocket engine installed, must accept a decrease in the useful load of the airplane at least equal to the weight of the standby power installation. This regulation provides a means of restoring this loss in recognition of the potential increase in airplane climb performance afforded by standby power.

An applicant for an increase in the maximum weight in accordance with this regulation is required to present an approved standby power installation and to furnish certain limitations and information in the form of placards or Airplane Flight Manual amendments.

Interested persons have been given an opportunity to comment on this regulation and consideration has been given to all relevant matter presented. In this connection, the notice of proposed rule making would have required a placard containing a precaution concerning the potential fire hazard of the hot standby power rocket engine casing. However, based upon some of the comments, the Agency has now determined that the casings of the rocket engines used for standby power do not, immediately following the firing, attain temperatures as high as had been anticipated. Therefore, there does not appear to be a po-

tential fire hazard with respect to such casings. For this reason, the proposed placard is not required in the final rule.

There were also comments objecting to the proposal insofar as it limited the increase in weight for multiengine airplanes to an amount which, together with the currently approved maximum weights, would equal the weight at which compliance is shown with the applicable one-engine-inoperative final takeoff or en route climb requirements with the standby power rocket engine inoperative. After further consideration of this provision, it now appears that there is a significant improvement in the one-engine-inoperative takeoff flight path due to thrust augmentation from the standby power, while at the same time there is only a small decrease in the subsequent one-engine-inoperative climb performance resulting from the weight and drag of the standby power rocket engine installation. Later stages of the en route operation are not considered critical since the progressive weight decrease of the airplane due to fuel burnoff compensates for the effects of drag created by the standby power rocket engine installation. Furthermore, service experience with airplanes which have been approved for an increased weight with a standby power installation does not indicate that the proposed limitation is necessary. For the foregoing reasons, the proposed limitation is not included in the final rule.

Another change being made concerns the weight which may be increased under this regulation. As proposed, there could have been an increase in the currently approved maximum takeoff and landing weights at which compliance with the applicable first minute and normal climb performance has been shown. However, the Agency now considers the requirement to be unnecessarily restrictive. For this reason, such a limitation is not contained in the final rule.

There was also a comment indicating that information should be made available to airport fire and rescue personnel concerning the precautions, if any, that should be taken with unused propellants in case of a fire. While this comment goes beyond the scope of this notice, it should be noted that on the basis of impact tests and fire tests with rocket engines, it does not appear that any special precautions would be necessary over and above those which would normally be exercised with respect to fires involving aircraft without standby power rocket engine installation. Tests in which rocket engines have been dropped from considerable heights above the ground have shown that the rocket engine propellant would neither explode nor ignite on impact. Other tests in which rocket engines have been placed in a bed of fire have shown that the propellant would ignite after the rocket engine was exposed to the flames for a relatively long period

of time, but the severity of the burning propellant was considered to be no greater than that of a gasoline fire. In addition, the burning propellant did not cause the rocket engine to explode or to become a projectile.

Certain of the comments expressed the opinion that the regulation should require a flight demonstration of the performance capability and safety of operation of the airplane at the increased maximum weight with standby power inoperative rather than confining the flight demonstration to a showing that the rocket engines and their controls can be operated safely and reliably at the increase in maximum weight. The Agency recognizes that both the drag of the inoperative rocket engine and the increased weight for the standby power installation will, to some extent, reduce the performance of the airplane in all flight regimes. However, experience with airplanes which have been granted limited weight credit for rocket engine installation does not indicate a significant reduction in performance or that the limited weight increase adversely affects safety. Furthermore, when the rocket engines are operated, the climb performance is significantly improved over that of the same type airplanes without rocket engines. This characteristic is ensured by the formula set forth in these amendments. Therefore, it is not considered that a flight demonstration is necessary. Should a unique installation be presented which has high drag and deleterious effects on performance, the required flight test of the rocket engine installation, which would involve a flight demonstration of the airplane, would give flight test personnel adequate opportunity to detect any unsafe condition.

A minimum "fail safe" requirement for all standby power installations in the form of dual rockets capable of being controlled and operated independently of each other was also suggested. While credit for the installation of more than one rocket engine would be permitted under this regulation, it is considered that such installation should be optional with the applicant. Adequate reliability will be assured for the rocket engine and its installation by the requirements that the rocket engine be type certificated, that the reliability of the installation be demonstrated, and that a safe-line limitation be observed for the rocket engine. The Agency does not believe that it is either appropriate or necessary in the interest of safety to require the installation of dual rockets.

A suggestion was made that the proposed regulation be changed to permit a weight increase of 2 percent above the maximum structural weight established for the airplane without standby power installed. In support of this suggestion, it was pointed out that fuel burnoff during taxiing, takeoff, and climb would compensate for a good part of the weight

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of the standby power installation before the airplane reaches its cruising speed. The Agency is aware that there may be airplanes which have a structural margin of at least 2 percent of their maximum structural weight. However, there may be others that were designed to the limits of the structural standards and the structural margin, if any, cannot be readily determined for each airplane. Furthermore, the temporary increase in climb performance which is available with standby power does not justify lowering the minimum structural standards.

Under Notice 63-41, the regulations concerning limited weight credit for airplanes equipped with standby power would have been set forth in a Special Civil Air Regulation. However, the regulations have subsequently been rescinded and it has been determined that the provisions of Notice 63-41 should be incorporated into the airworthiness parts (Parts 23 and 25) of the Federal Aviation Regulations rather than in a special regulation. It is also considered appropriate to simultaneously incorporate the requirements of Special Federal Aviation Regulation SFAR-14 (formerly SR-426), "Performance Credit For Transport Category Airplanes Equipped With Standby Power", into Part 25 of the Federal Aviation Regulations. Since this action merely continues an existing regulation without substantive change and imposes no additional burden on any person, notice and public procedure thereon are unnecessary.

The result of incorporating the proposal into Part 23 of the FARs would be to exclude from its coverage those aircraft certificated under Part 4a of the Civil Air Regulations. However, the Agency has determined that there is a possibility that the operators of such aircraft may wish to take advantage of the increased weights authorized by this Amendment. Therefore a paragraph has been added to Appendix E of Part 23 to the effect that Part 4a aircraft may, for the purposes of increased weights due to standby power, be treated as if they had been certificated under Part 3 of the Civil Air Regulations or Part 23 of the Federal Aviation Regulations. This will permit operators of Part 4a aircraft to avail themselves of the privileges of this regulation to the same extent as operators of Part 3 (CAR) or Part 23 (FAR) aircraft.

The regulations covered by Notice 63-41 and the requirements of former SFAR-14 are both being set forth in new Appendixes to Parts 23 and 25. Appropriate references to these Appendixes have been made in the provisions of Parts 23 and 25 concerning the limitations on maximum weights in order to permit the continued use of the provisions of former SFAR-14 in determining maximum weights and to permit limited increases in such weights as proposed in Notice 63-41.

These amendments are made under the authority of sections 313(a), 601, 603, 604, and 605 of the Federal Aviation Act of 1958 (49 U.S.C. 1354(a), 1421, 1423, 1424, 1425).

In consideration of the foregoing, SFAR-14 is hereby rescinded and Parts 23 and 25 of the Federal Aviation Reg-

ulations (14 CFR 23 and 25), are amended as follows, effective August 1, 1965:

1. Section 23.25(a)(1)(iii) is amended to read as follows:

§ 23.25 Weight limits.

(a) *Maximum weight.* * * *

(1) * * *

(iii) The highest weight at which compliance with each applicable flight requirement is shown, except for airplanes equipped with standby power rocket engines, in which case it is the highest weight established in accordance with Appendix E of this part.

2. Part 23 is amended by adding the following new appendix at the end thereof:

Appendix E—Limited Weight Credit For Airplanes Equipped With Standby Power

(a) Each applicant for an increase in the maximum certificated takeoff and landing weights of an airplane equipped with a type-certificated standby power rocket engine may obtain an increase as specified in paragraph (b) if—

(1) The installation of the rocket engine has been approved and it has been established by flight test that the rocket engine and its controls can be operated safely and reliably at the increase in maximum weight; and

(2) The Airplane Flight Manual, or the placard, markings or manuals required in place thereof, set forth in addition to any other operating limitations the Administrator may require, the increased weight approved under this regulation and a prohibition against the operation of the airplane at the approved increased weight when—

(i) The installed standby power rocket engines have been stored or installed in excess of the time limit established by the manufacturer of the rocket engine (usually stenciled on the engine casing); or

(ii) The rocket engine fuel has been expended or discharged.

(b) The currently approved maximum takeoff and landing weights at which an airplane is certificated without a standby power rocket engine installation may be increased by an amount which does not exceed any of the following:

(1) An amount equal in pounds to 0.014 IN, where I is the maximum usable impulse in pounds-seconds available from each standby power rocket engine and N is the number of rocket engines installed.

(2) An amount equal to 5 percent of the maximum certificated weight approved in accordance with the applicable airworthiness regulations without standby power rocket engines installed.

(3) An amount equal to the weight of the rocket engine installation.

(4) An amount that, together with the currently approved maximum weight, would equal the maximum structural weight established for the airplane without standby power rocket engines installed.

(c) For the purposes of this Appendix, "standby power" is power or thrust, or both, obtained from rocket engines for a relatively short period and actuated only in cases of emergency.

(d) For the purposes of limited weight credit for airplanes equipped with standby power, as set forth in § 23.25(a)(1)(iii) and this Appendix, an airplane certificated under Part 4a of the Civil Air Regulations is treated as if it had been certificated under Part 3 of the Civil Air Regulations or Part 23 of the Federal Aviation Regulations.

3. Section 25.25(a)(3) is amended to read as follows:

§ 25.25 Weight limits.

(a) *Maximum weight.* * * *

(3) The highest weight at which compliance with each applicable flight requirement is shown, except for airplanes equipped with standby power rocket engines, in which case it is the highest weight established in accordance with Appendix E of this part.

4. Section 25.59 is amended by adding the following new paragraph at the end thereof:

§ 25.59 Takeoff path.

(c) For airplanes equipped with standby power rocket engines, the takeoff path may be determined in accordance with section II of Appendix E.

5. Section 25.111 is amended by adding the following new paragraph at the end thereof:

§ 25.111 Takeoff path.

(e) For airplanes equipped with standby power rocket engines, the takeoff path may be determined in accordance with section II of Appendix E.

6. Part 25 is amended by adding the following new Appendix at the end thereof:

Appendix E

I—Limited Weight Credit For Airplanes Equipped With Standby Power

(a) Each applicant for an increase in the maximum certificated takeoff and landing weights of an airplane equipped with a type-certificated standby power rocket engine may obtain an increase as specified in paragraph (b) if—

(1) The installation of the rocket engine has been approved and it has been established by flight test that the rocket engine and its controls can be operated safely and reliably at the increase in maximum weight; and

(2) The Airplane Flight Manual, or the placard, markings or manuals required in place thereof, set forth in addition to any other operating limitations the Administrator may require, the increased weight approved under this regulation and a prohibition against the operation of the airplane at the approved increased weight when—

(i) The installed standby power rocket engines have been stored or installed in excess of the time limit established by the manufacturer of the rocket engine (usually stenciled on the engine casing); or

(ii) The rocket engine fuel has been expended or discharged.

(b) The currently approved maximum takeoff and landing weights at which an airplane is certificated without a standby power rocket engine installation may be increased by an amount that does not exceed any of the following:

(1) An amount equal in pounds to 0.014 IN, where I is the maximum usable impulse in pounds-seconds available from each standby power rocket engine and N is the number of rocket engines installed.

(2) An amount equal to 5 percent of the maximum certificated weight approved in accordance with the applicable airworthiness regulations without standby power rocket engines installed.

(3) An amount equal to the weight of the rocket engine installation.

(4) An amount that, together with the currently approved maximum weight, would equal the maximum structural weight established for the airplane without standby power rocket engines installed.

II--Performance Credit for Transport Category Airplanes Equipped With Standby Power

The Administrator may grant performance credit for the use of standby power on transport category airplanes. However, the performance credit applies only to the maximum certificated takeoff and landing weights, the takeoff distance, and the takeoff paths, and may not exceed that found by the Administrator to result in an overall level of safety in the takeoff, approach, and landing regimes of flight equivalent to that prescribed in the regulations under which the airplane was originally certificated without standby power. For the purposes of this Appendix, "standby power" is power or thrust, or both, obtained from rocket engines for a relatively short period and actuated only in cases of emergency. The following provisions apply:

(1) *Takeoff; general.* The takeoff data prescribed in §§ (2) and (3) must be determined at all weights and altitudes, and at ambient temperatures if applicable, at which performance credit is to be applied.

(2) *Takeoff path.*

(a) The one-engine-inoperative takeoff path with standby power in use must be determined in accordance with the performance requirements of the applicable airworthiness regulations.

(b) The one-engine-inoperative takeoff path (excluding that part where the airplane is on or just above the takeoff surface) determined in accordance with paragraph (a) of this section must lie above the one-engine-inoperative takeoff path without standby power at the maximum takeoff weight at which all of the applicable airworthiness requirements are met. For the purpose of this comparison, the flight path is considered to extend to at least a height of 400 feet above the takeoff surface.

(c) The takeoff path with all engines operating, but without the use of standby power, must reflect a conservatively greater overall level of performance than the one-engine-inoperative takeoff path established in accordance with paragraph (a) of this section. The margin must be established by the Administrator to insure safe day-to-day operations, but in no case may it be less than 15 percent. The all-engines-operating take-

off path must be determined by a procedure consistent with that established in complying with paragraph (a) of this section.

(d) For reciprocating-engine-powered airplanes, the takeoff path to be scheduled in the Airplane Flight Manual must represent the one-engine-inoperative takeoff path determined in accordance with paragraph (a) of this section and modified to reflect the procedure (see § (6)) established by the applicant for flap retraction and attainment of the en route speed. The scheduled takeoff path must have a positive slope at all points of the airborne portion and at no point must it lie above the takeoff path specified in paragraph (a) of this section.

(3) *Takeoff distance.* The takeoff distance must be the horizontal distance along the one-engine-inoperative takeoff path determined in accordance with § (2) (a) from the start of the takeoff to the point where the airplane attains a height of 50 feet above the takeoff surface for reciprocating-engine-powered airplanes and a height of 35 feet above the takeoff surface for turbine-powered airplanes.

(4) *Maximum certificated takeoff weights.* The maximum certificated takeoff weights must be determined at all altitudes, and at ambient temperatures, if applicable, at which performance credit is to be applied and may not exceed the weights established in compliance with paragraphs (a) and (b) of this section.

(a) The conditions of § (2) (b) through (d) must be met at the maximum certificated takeoff weight.

(b) Without the use of standby power, the airplane must meet all of the en route requirements of the applicable airworthiness regulations under which the airplane was originally certificated. In addition, turbine-powered airplanes without the use of standby power must meet the final takeoff climb requirements prescribed in the applicable airworthiness regulations.

(5) *Maximum certificated landing weights.* (a) The maximum certificated landing weights (one-engine-inoperative approach and all-engines-operating landing climb) must be determined at all altitudes, and at ambient temperatures if applicable, at which performance credit is to be applied and must not exceed that established in compliance with paragraph (b) of this section.

(b) The flight path, with the engines operating at the power or thrust, or both, appropriate to the airplane configuration and with standby power in use, must lie above the flight path without standby power in use at the maximum weight at which all of the applicable airworthiness requirements are met. In addition, the flight paths must comply with subparagraphs (i) and (ii) of this paragraph.

(i) The flight paths must be established without changing the appropriate airplane configuration.

(ii) The flight paths must be carried out for a minimum height of 400 feet above the point where standby power is actuated.

(6) *Airplane configuration, speed, and power and thrust; general.* Any change in the airplane's configuration, speed, and power or thrust, or both, must be made in accordance with the procedures established by the applicant for the operation of the airplane in service and must comply with paragraphs (a) through (c) of this section. In addition, procedures must be established for the execution of balked landings and missed approaches.

(a) The Administrator must find that the procedure can be consistently executed in service by crews of average skill.

(b) The procedure may not involve methods or the use of devices which have not been proven to be safe and reliable.

(c) Allowances must be made for such time delays in the execution of the procedures as may be reasonably expected to occur during service.

(7) *Installation and operation; standby power.* The standby power unit and its installation must comply with paragraphs (a) and (b) of this section.

(a) The standby power unit and its installation must not adversely affect the safety of the airplane.

(b) The operation of the standby power unit and its control must have proven to be safe and reliable.

Issued in Washington, D.C., on June 24, 1965.

N. E. HALABY,
Administrator.